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Image Editing with Diffusion Models

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Neural Networks

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Task

This project follows the approach introduced in “*InstructPix2Pix: Learning to Follow Image Editing Instructions*”

Main steps:

- Generating instructions and paired captions
- Generating paired images from paired Captions
- Fine-tuning Stable Diffusion (SD)

InstructPix2Pix: Learning to Follow Image Editing Instructions

Tim Brooks* Aleksander Holynski* Alexei A. Efros

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Figure 1. Given an image and an instruction for how to edit that image, our model performs the appropriate edit. Our model does not require full descriptions for the input or output image, and edits images in the forward pass without per-example inversion or fine-tuning.

Generating Instructions and Paired Captions

We fine-tuned Gemma 2B to generate an edit prompt and an edited prompt given an original prompt

Implementation details

- GPU: 40GB A100
- dataset size: 800
- epochs: 3
- batch size: 4
- grad. acc. steps: 2

Input:

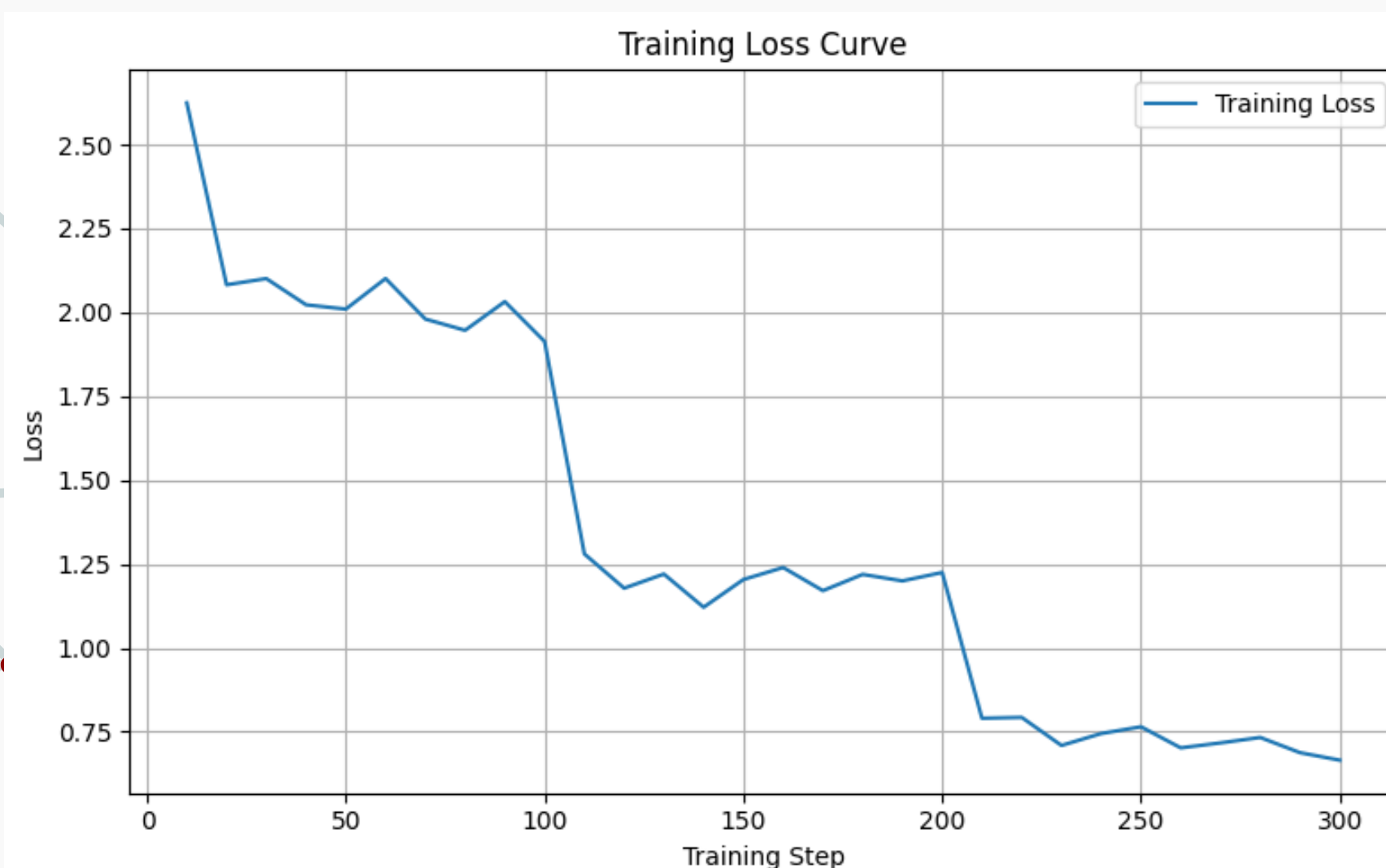
original prompt:
"A squirrel eating a burger"



Gemma 2B

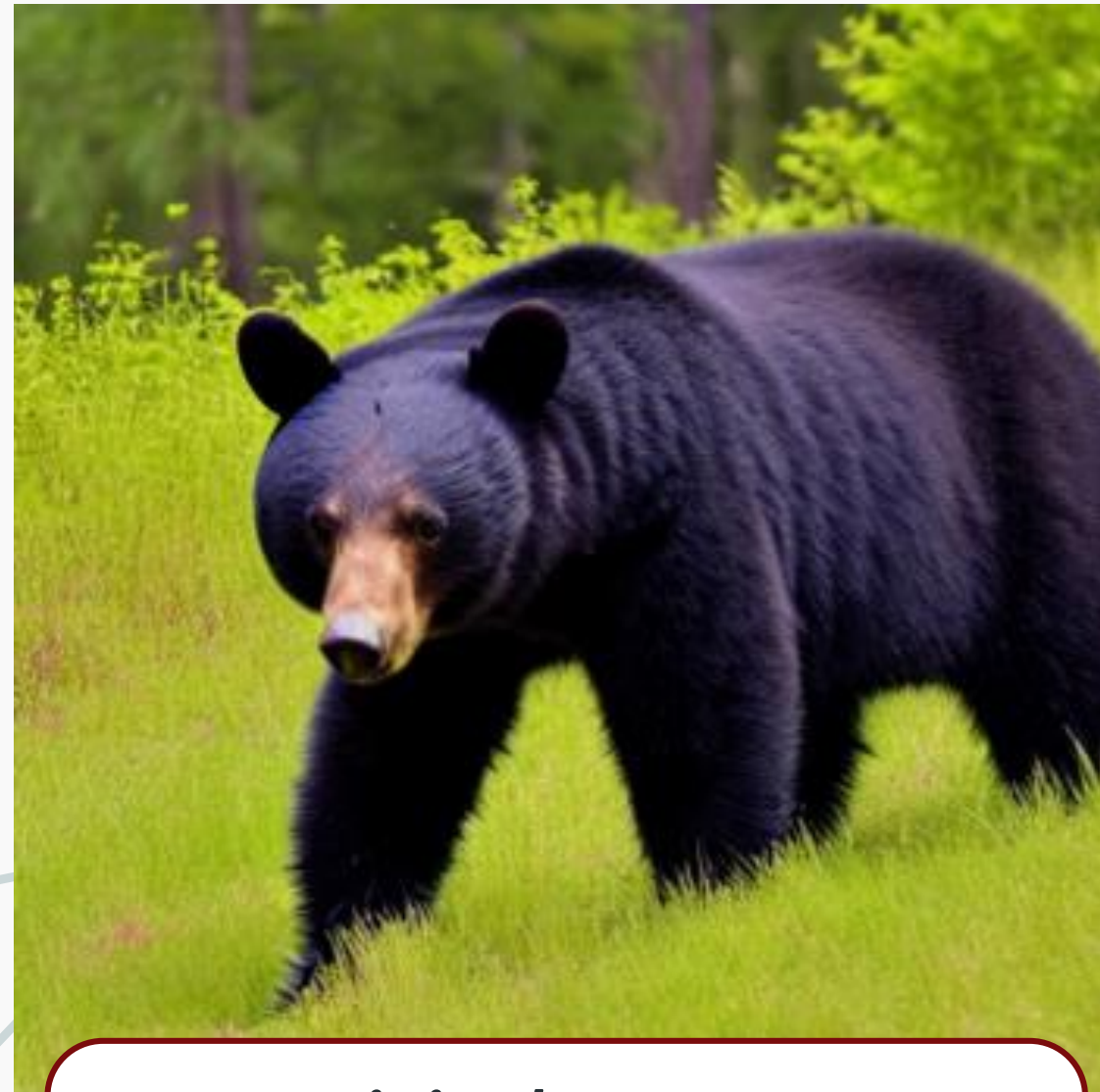
Output:

edit prompt:
*"Replace the
squirrel with a lion"*
+
edited prompt:
*"A lion eating a
burger"*



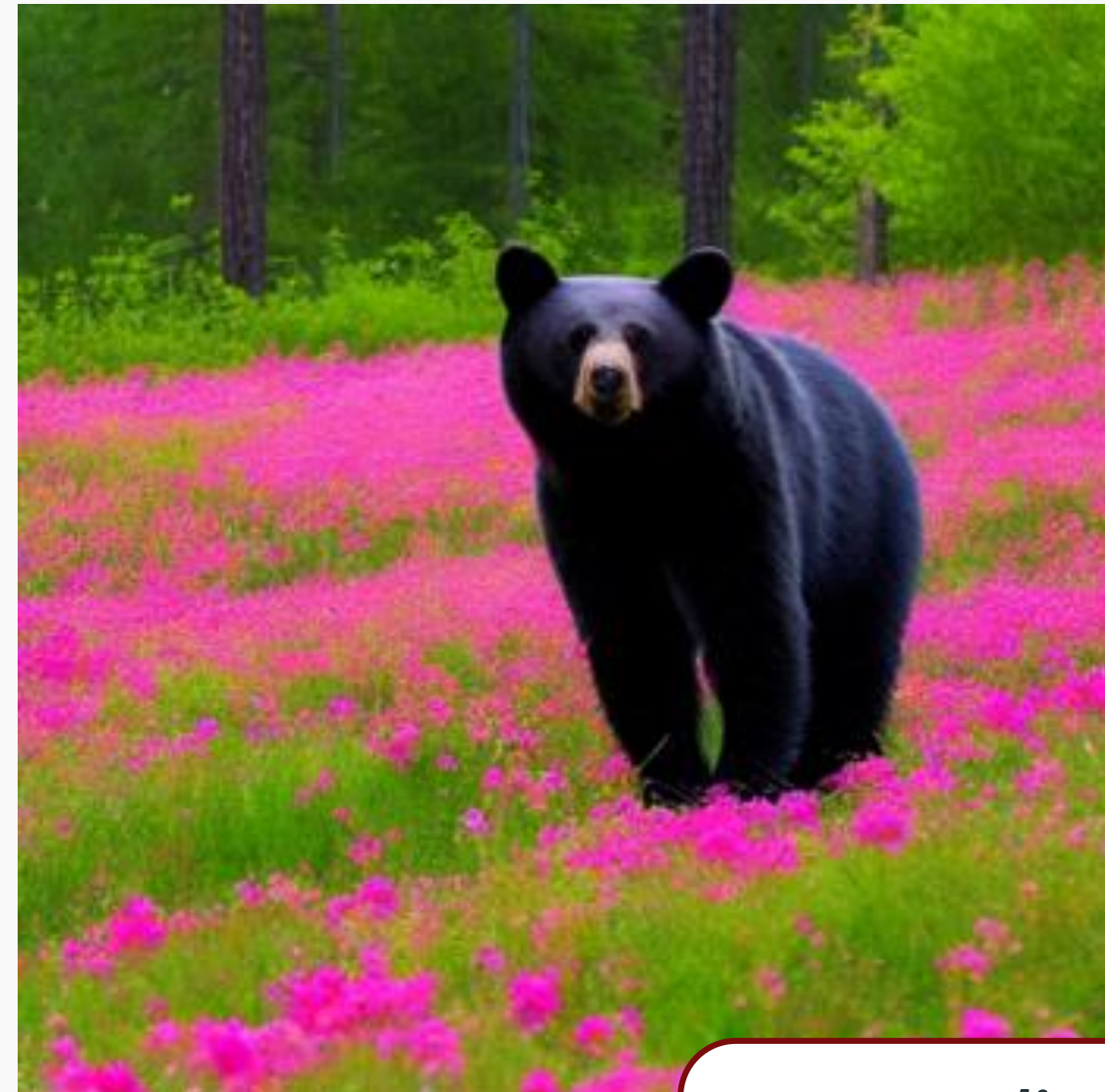
Generating Paired Images from Paired Captions

Stable Diffusion



original prompt:
“a black bear is walking in
the **grass** in the forest”

w/o Prompt2Prompt



edited prompt:
“a black bear is walking in
the **flowers** in the forest”

w/ Prompt2Prompt



*alpha=0.8
replace_ratio=0.9
weight=1.5*

w/ Prompt2Prompt



*alpha=0.4
replace_ratio=0.8
weight=5*

...but Prompt-to-Prompt requires tuning **several hyperparameters** (e.g., alpha, replace_ratio, weight, etc.).



Generating Paired Images from Paired Captions

Stable Diffusion



original prompt:
“a black bear is walking in
the *grass* in the forest”

w/o ControlNet



edited prompt:
“a black bear is walking in
the *flowers* in the forest”

w/ ControlNet



scale=0.10

w/ ControlNet



scale=0.15

original
prompt

Stable
Diffusion

original
image

Canny Edge

ControlNet

edited
prompt

Image Filtering Metrics:

- dir. similarity (>0.15)
- img–img similarity (>0.65)
- img–cap similarity (>0.20)

edited
image



Fine-tuning of Stable Diffusion

We fine-tuned **only the U-Net** of the SD to follow textual editing instructions

Implementation details

- GPU: 40GB A100
- dataset size: 1001
- resolution: 256x256
- batch size: 32
- learning rate: $5e-5$
- grad. acc. steps: 4
- max grad. norm: 1
- steps: 300

Input:

original image:



+ edit prompt:
"Make her a man"

Output:

edited image:

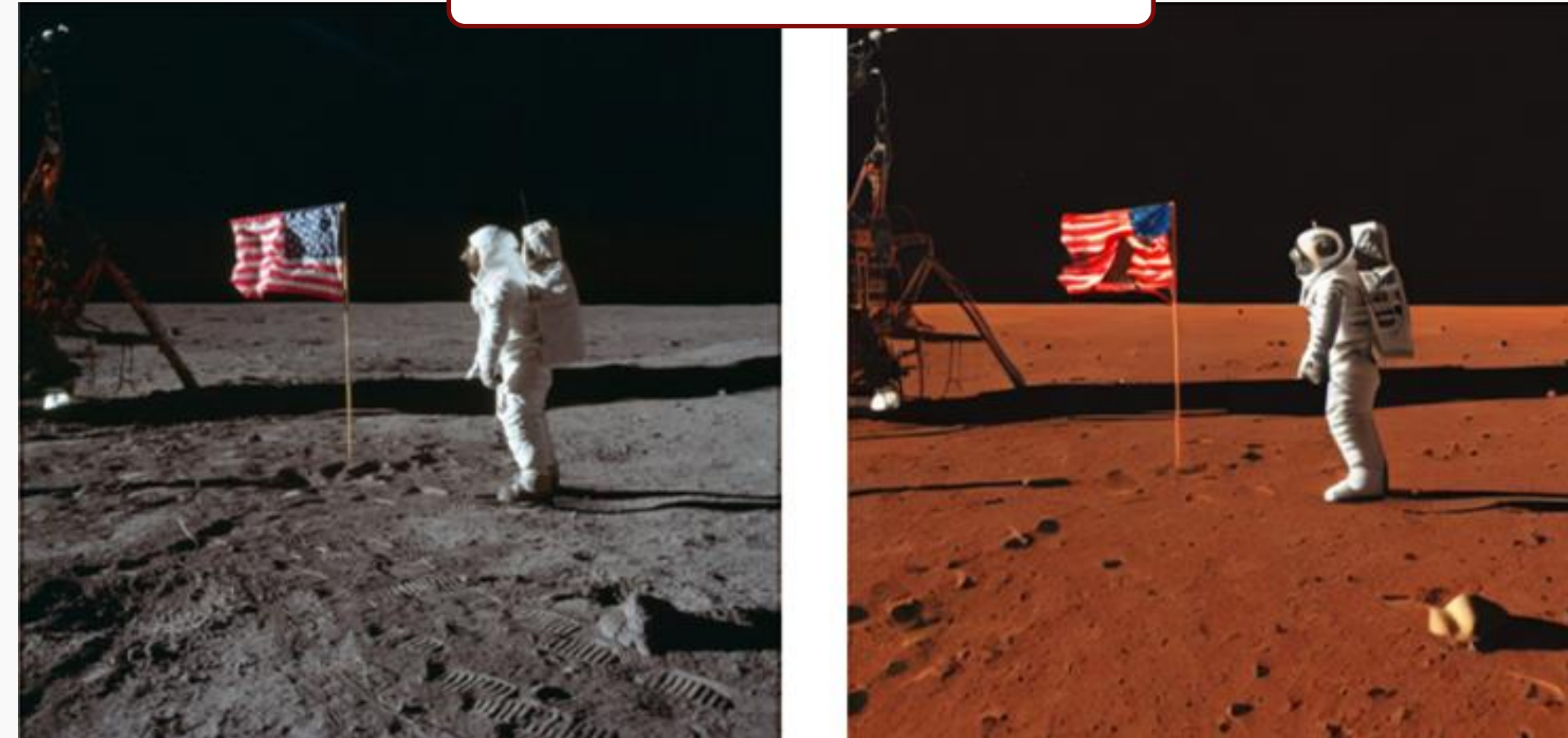


Results

“make him old”



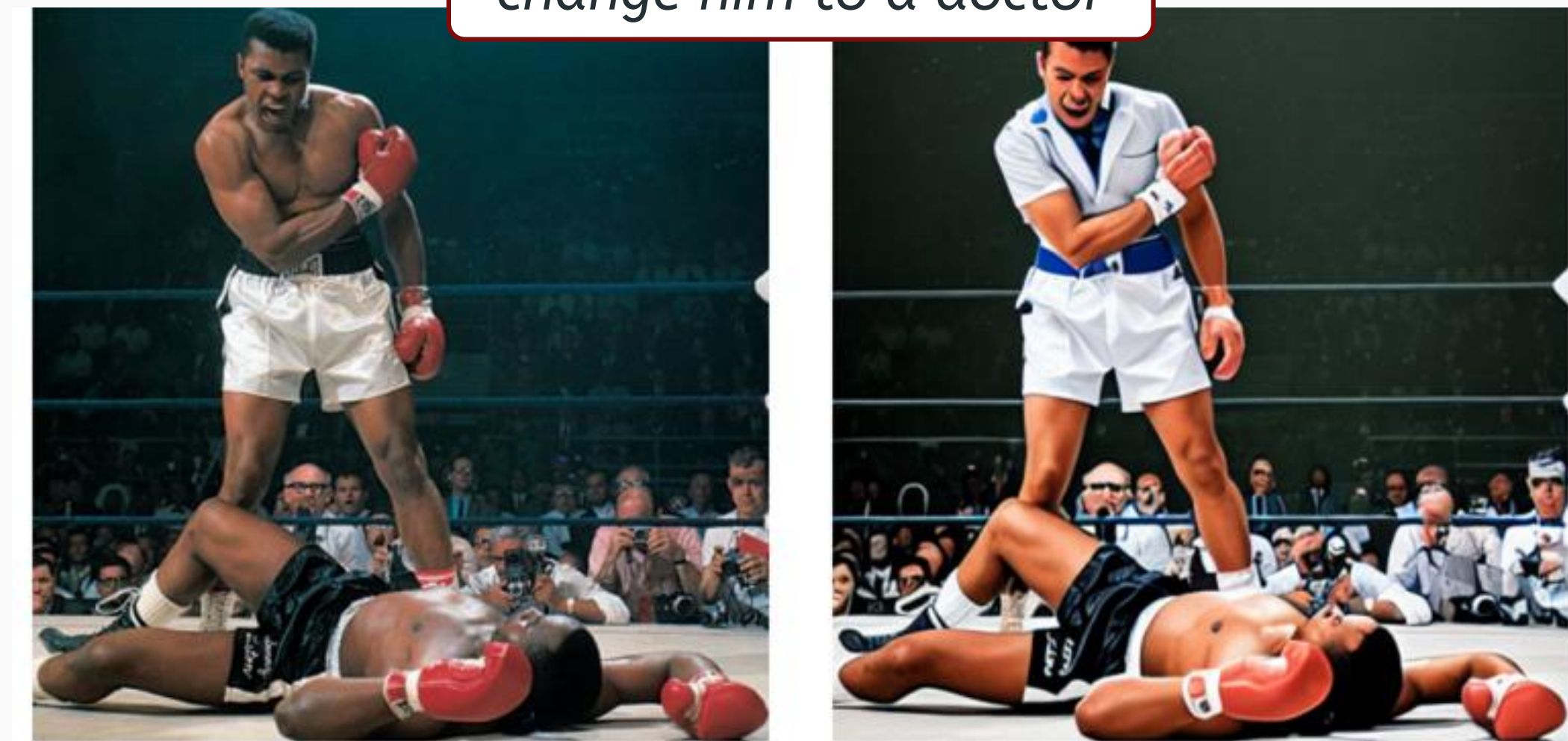
“move it to Mars”



“add glasses”



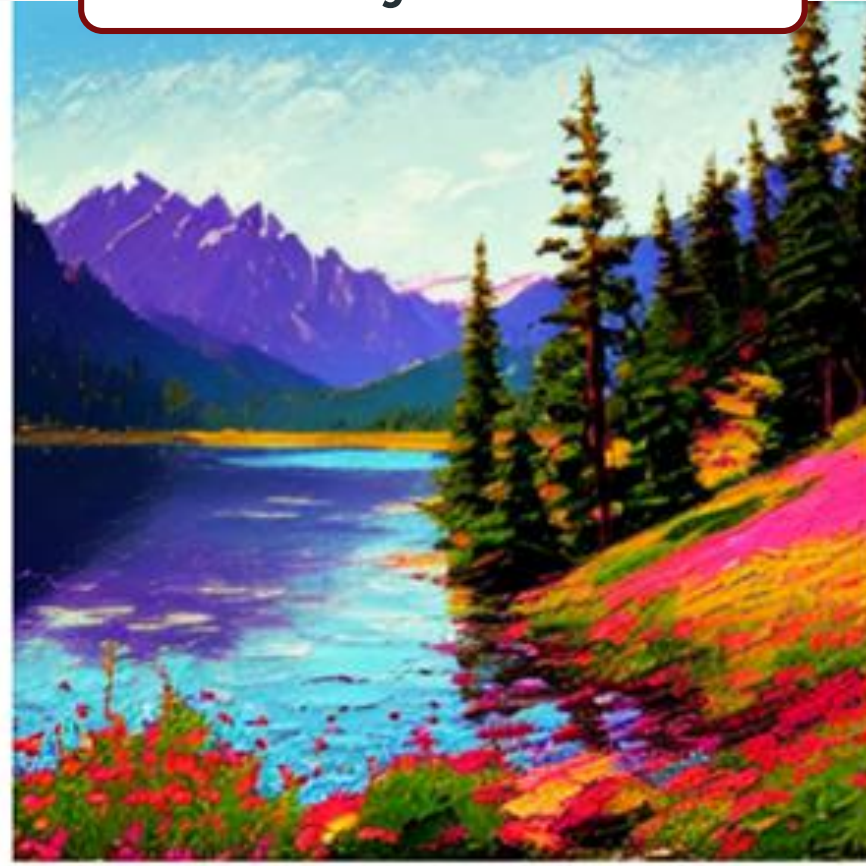
“change him to a doctor”



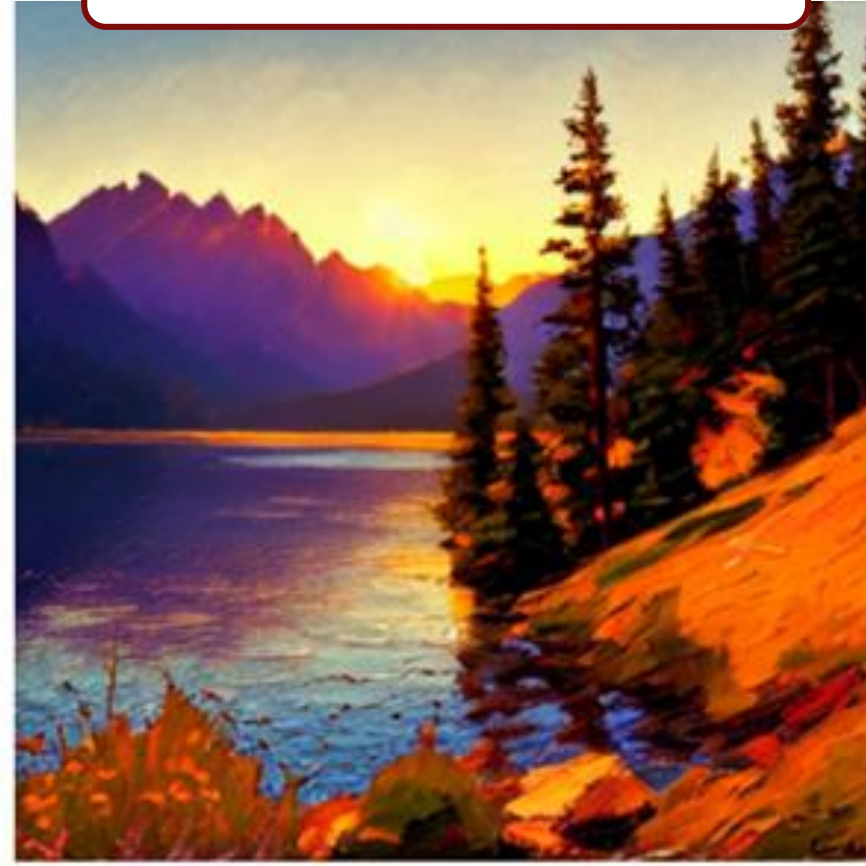
Results

Different contextual edits

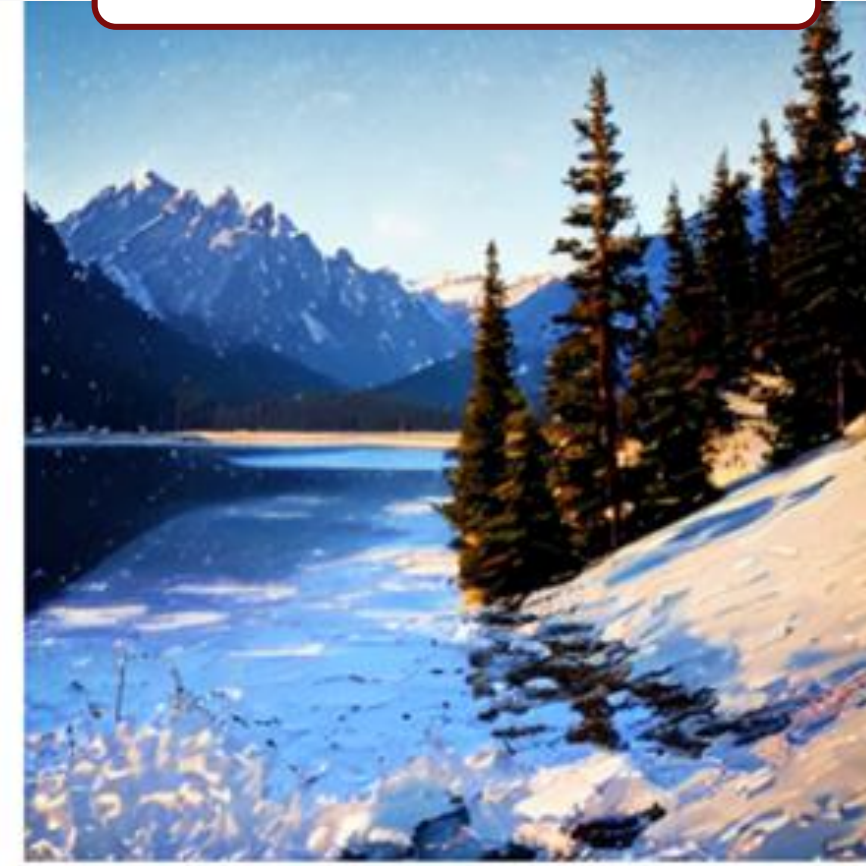
“add flowers”



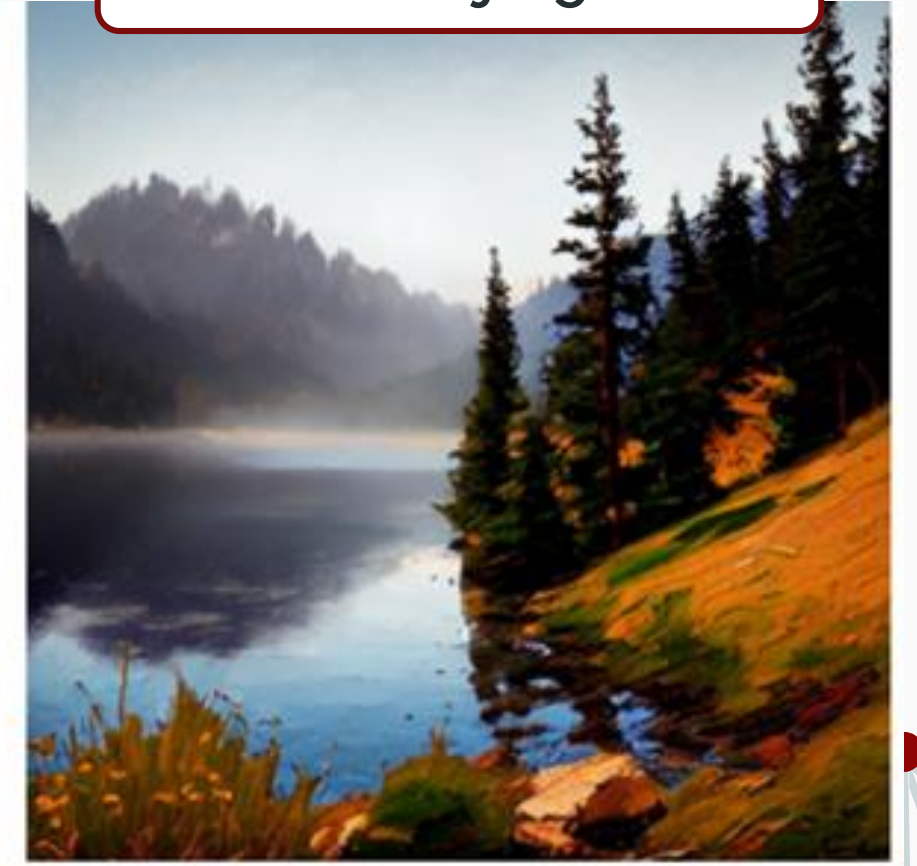
“at sunset”



“add snow”



“add fog”



“change to a polar bear”



“make him a wooden statue”



“turn it purple”



“as a cartoon”

Results

Classifier-free guidance weights over two conditional inputs



“turn into a cyborg”



Results

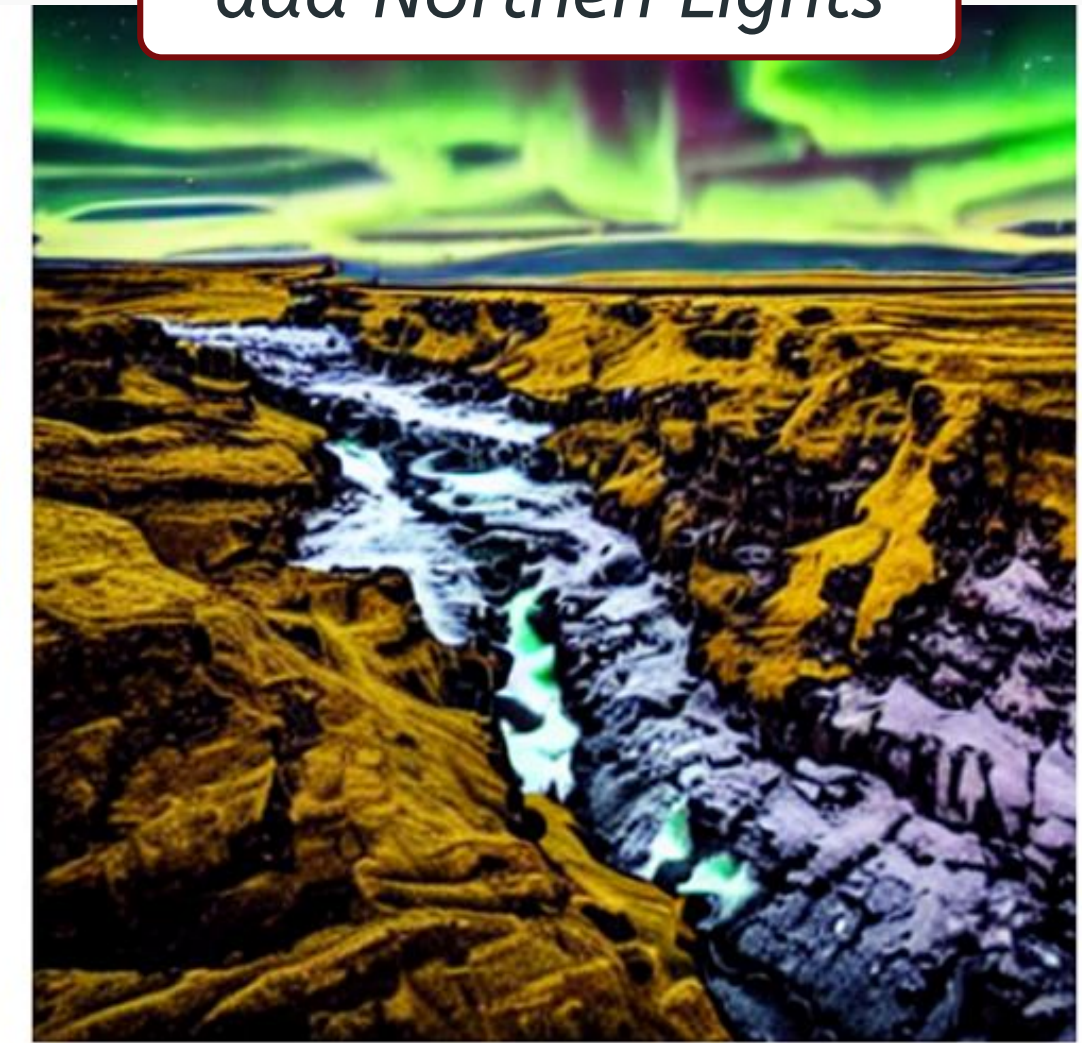
Sequential
Edits



“move to Iceland”



“add Northern Lights”



“she’s a nurse”



Different latent
noise

Conclusions

Limitations:

- The **dataset quality** is limited by the ability of:
 - Gemma 2B to generate instructions
 - SD to generate images
- **Limited fine-tuning data**
- **Resolution constraints**
- Difficulty with **complex instructions**
- **Accumulated errors** in sequential edits
- **Biases**

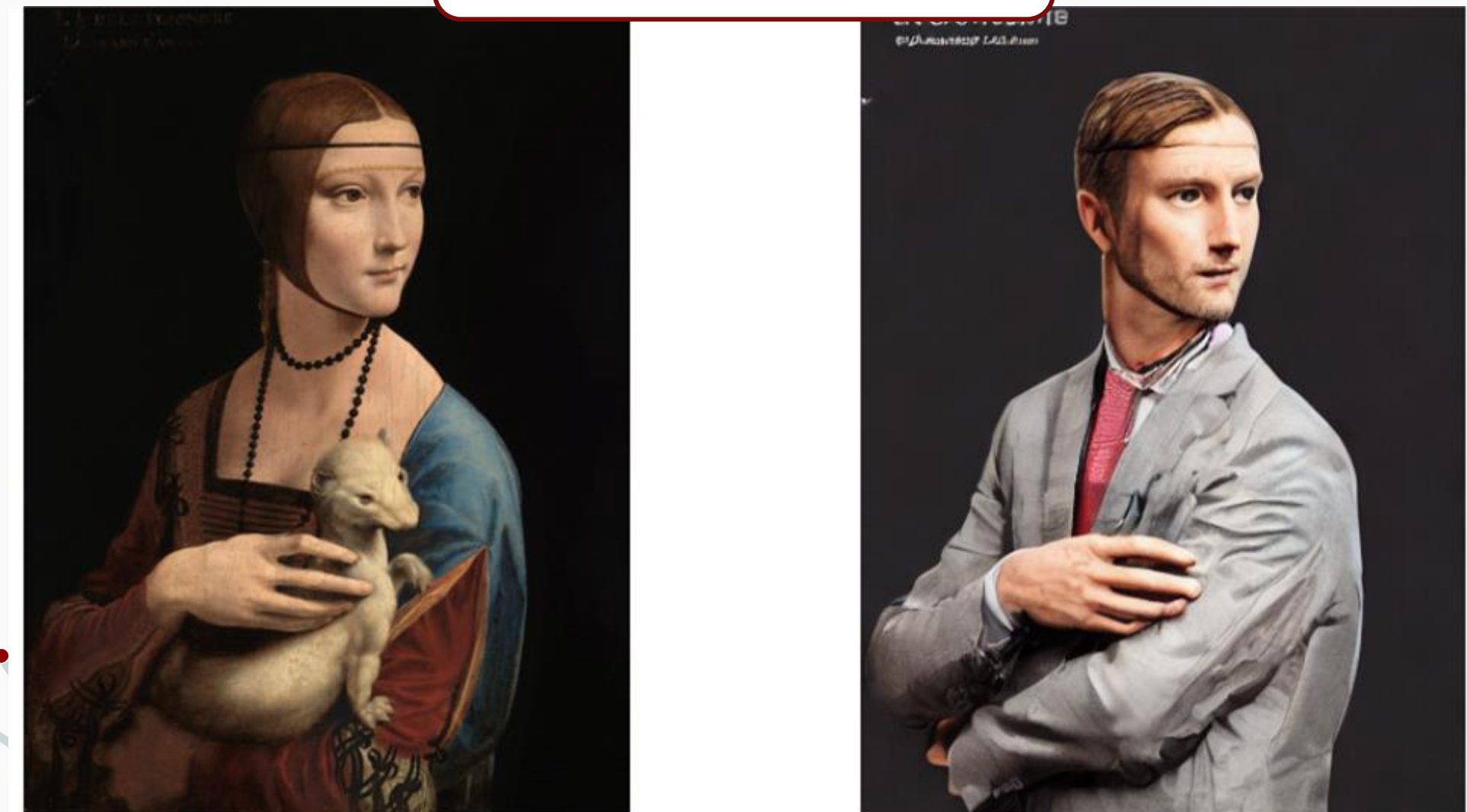
Future works:

- Use **newer SD versions** for dataset generation
- Fine-tune a **newer SD versions** to follow instructions
- Fine-tune on **larger, high-quality data**
- Address Biases
- Enhance robustness to complex instructions

“turn into a doctor”



“turn into a CEO”





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Thanks for your attention



Daniele
Sabatini



“after the university”



Luca
Conti



“after graduated”

References:

1. Tim Brooks, Aleksander Holynski, Alexei A. Efros. InstructPix2Pix: Learning to Follow Image Editing Instructions. arXiv:2211.09800, 2022.
<https://arxiv.org/abs/2211.09800>
2. Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, Daniel Cohen-Or. Prompt-to-Prompt Image Editing with Cross Attention Control. arXiv:2208.01626, 2022. <https://arxiv.org/abs/2208.01626>
3. Lvmin Zhang, Maneesh Agrawala. Adding Conditional Control to Text-to-Image Diffusion Models. arXiv:2302.05543, 2023. <https://arxiv.org/abs/2302.05543>
4. Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, Björn Ommer. [High-Resolution Image Synthesis with Latent Diffusion Models](#). CVPR 2022.